

Lecture 4

Liouville integrability and the Liouville–Arnold theorem

Objectives. *The central theorem of Part I. On a $2n$ -dimensional phase space, n independent integrals in involution force the motion onto an n -dimensional torus, and on that torus the motion is a straight line traversed at constant speed. We state the theorem with every hypothesis visible, prove it in six stages, say at which stage each hypothesis is spent, and read off what the motion on a torus looks like. A final section, for reading, shows that the theorem is only local: the tori can be glued together in a twisted way.*

4.1 Independence and involution

Everything in this lecture is built from three results already proved.

- $X_g f = \{f, g\}$, from $X_g h = \{h, g\}$ (Definition 1.14); so $\{f, g\} = 0$ says that f is constant along the flow of g .
- $\{F_i, F_j\} = 0$ implies $[X_{F_i}, X_{F_j}] = 0$, and then, if the fields are complete, their flows commute (Corollary 1.16).
- $\omega(X_f, X_g) = \{f, g\}$ (Proposition 3.6), and at most n independent functions can be in involution (Proposition 3.22).

A *first integral* of H is a function F with $\{F, H\} = 0$ (Definition 1.13). Each one confines the motion to a level set of F — near any point where $dF \neq 0$, a hypersurface of dimension $2n - 1$. With several first integrals $F_1, \dots, F_k$ we collect them into one map,

$$F = (F_1, \dots, F_k) : \mathcal{M} \rightarrow \mathbb{R}^k, \quad \mathcal{M}_f = F^{-1}(f) = \{x : F_i(x) = f_i \text{ for all } i\}, \quad (4.1)$$

and the motion stays in the level set $\mathcal{M}_f$ through its initial point. Two things can go wrong with such a collection, and each has a name.

Definition 4.1 (Independence). $F_1, \dots, F_k$ are *independent at x* if $dF_1, \dots, dF_k$ are linearly independent at x — equivalently, if the $k \times 2n$ Jacobian matrix of F has rank k there. They are *independent* if this holds on a dense open subset of $\mathcal{M}$.

Independence at a point is what the implicit function theorem needs: if it holds at $x \in \mathcal{M}_f$, then near x the level set $\mathcal{M}_f$ is a smooth submanifold of dimension $2n - k$, with tangent space

$$T_x \mathcal{M}_f = \{v : dF_i(v) = 0 \text{ for all } i\}. \quad (4.2)$$

Asking for it only on a dense open set is deliberate: in every example it fails somewhere, and the places where it fails are the interesting ones.

Definition 4.2 (Involution). $F_1, \dots, F_k$ are in involution if $\{F_i, F_j\} = 0$ for all i, j .

Example 4.3 (The two-dimensional oscillator). On $\mathbb{R}^4$ take $H = H_1 + H_2$ with $H_i = \frac{1}{2}(p_i^2 + \nu_i^2 q_i^2)$, $\nu_i > 0$. The H_i are in involution, since they involve disjoint variables. Their Jacobian is

$$\begin{pmatrix} \nu_1^2 q_1 & 0 & p_1 & 0 \\ 0 & \nu_2^2 q_2 & 0 & p_2 \end{pmatrix} \quad (\text{columns } q_1, q_2, p_1, p_2),$$

of rank 2 unless one row vanishes, that is unless $q_1 = p_1 = 0$ or $q_2 = p_2 = 0$. Those are two planes in $\mathbb{R}^4$, closed with empty interior, so the H_i are independent. The level sets are pictured in Figure 4.1: the image of $F = (H_1, H_2)$ is the closed quadrant, over each interior point sits a torus, and over the edges — exactly where independence fails — the torus collapses to a circle or a point.

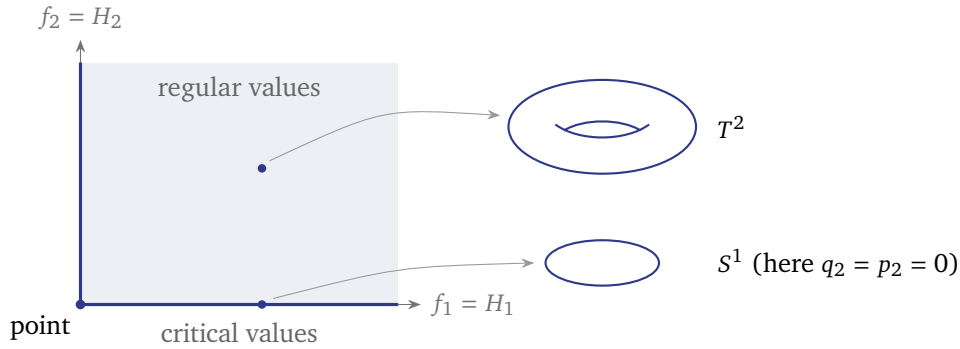


Figure 4.1: The level sets of $F = (H_1, H_2)$ for the two-dimensional oscillator foliate $\mathbb{R}^4$. Over an interior point (f_1, f_2) the level set is the torus $\{p_1^2 + \nu_1^2 q_1^2 = 2f_1\} \times \{p_2^2 + \nu_2^2 q_2^2 = 2f_2\}$, a circle in each symplectic plane. On the edges of the quadrant, where the rank of dF drops, one circle has shrunk to a point; at the corner both have. This is the picture the Liouville–Arnold theorem generalises.

Each of the two conditions is easy to have without the other.

Example 4.4 (Independent but not in involution). For a particle in a central potential on $\mathbb{R}^3$, $H = \frac{1}{2}|p|^2 + V(|q|)$, the three components of $\ell = q \times p$ are first integrals. So H, ℓ_1, ℓ_2 are three first integrals on a 6-dimensional phase space, and they are independent. They are not in involution:

$$\{\ell_1, \ell_2\} = \ell_3, \tag{4.3}$$

and cyclically. Proposition 3.22 says three commuting independent functions is the most $n = 3$ allows; three non-commuting ones are not the same thing, and § 4.3 shows what goes wrong with them.

Example 4.5 (In involution but not independent). For any F and any function g of one variable, $\{F, g(F)\} = g'(F)\{F, F\} = 0$. Involution is free for functions of one another, so it carries information only together with independence.

4.2 Liouville integrable systems: the algebraic definition

Definition 4.6 (Liouville integrable). A Hamiltonian system (M, ω, H) with $\dim M = 2n$ is *Liouville integrable* if there are n independent functions $F_1, \dots, F_n$ in involution such that H is a function of them, $H = h(F_1, \dots, F_n)$. Most often H is one of them, $H = F_1$.

By [Proposition 3.22](#), n is the most one can ask for. Nothing in the definition mentions compactness, and that is not an oversight: it is an *algebraic* condition on a list of functions, checked by computing brackets and a rank. [Lecture 9](#) will verify it for the open Toda chain, whose level sets are not compact, and the theorem of the next section will then not apply as stated.

Two remarks on the definition. First, $H = h(F)$ is automatically a first integral of each F_i , since it is built from functions that commute with F_i ; and by the chain rule $dH = \sum_i \partial_i h dF_i$, so

$$X_{h(F)} = \sum_i \frac{\partial h}{\partial f_i}(F) X_{F_i}. \quad (4.4)$$

Second, the functions are part of the data but not canonical: (H_1, H_2) and (H, H_2) in [Example 4.3](#) define the same level sets.

Example 4.7 (Integrable systems we already own). 1. $n = 1$. Every H with $dH \neq 0$ on a dense open set.

2. *The n -dimensional oscillator*, with $F_i = H_i$ as in [Example 4.3](#).
3. *A central force in $\mathbb{R}^3$* , with $F = (H, |\ell|^2, \ell_3)$. Involution: $\{H, \ell_i\} = 0$ for every i , and $\{|\ell|^2, \ell_3\} = 0$ by (4.3). Independence: the p -components of $d\ell_i$ are $e_i \times q$, all orthogonal to q , while that of dH is p ; so dH is independent of the other two wherever $q \cdot p \neq 0$, and $d|\ell|^2 = 2 \sum_i \ell_i d\ell_i$ is independent of $d\ell_3$ wherever ℓ is not parallel to e_3 . Both conditions fail only on the zero set of a nonzero polynomial.
4. *The free rigid body with distinct principal moments of inertia*, on $T^*SO(3)$, with H , the squared length of the spatial angular momentum and one of its components — the same pattern as item 3, stated here without the computation. For a spherical body H is itself a multiple of the squared length, and the triple is dependent.

4.3 The theorem, hypothesis by hypothesis

Theorem 4.8 (Liouville–Arnold · proved modulo [Lemma 4.11](#) and [Theorem 1.10](#)). Let (M, ω) be a symplectic manifold of dimension $2n$, and $F_1, \dots, F_n$ functions in involution. Fix $f \in \mathbb{R}^n$, let Σ be a connected component of the level set M_f , and suppose

- (I) $dF_1, \dots, dF_n$ are linearly independent at every point of Σ ; and
- (C) Σ is compact.

Then:

1. Σ is a smooth Lagrangian submanifold, invariant under the flow of every X_{F_i} ;
2. Σ is diffeomorphic to the torus $T^n = \mathbb{R}^n / 2\pi\mathbb{Z}^n$, with angle coordinates $\varphi_1, \dots, \varphi_n$ defined modulo 2π ;
3. for any $H = h(F)$, the motion on Σ is $\varphi(t) = \varphi(0) + \nu t$, with a frequency vector $\nu \in \mathbb{R}^n$ that depends on f only.

Two hypotheses are named and two more are hidden in the words. Each one is needed, and the cleanest way to see it is to delete it.

- *Involution.* On $\mathbb{R}^4$ take $F_1 = \frac{1}{2}(q_1^2 + p_1^2 + q_2^2 + p_2^2)$ and $F_2 = q_1$. For $f_1 > f_2^2/2$ the level set $\{q_1 = f_2, p_1^2 + q_2^2 + p_2^2 = 2f_1 - f_2^2\}$ is a 2-sphere: compact, connected, and regular. But $\{F_1, F_2\} = -p_1$, the flow of F_2 changes F_1 ($X_{F_2}F_1 = \{F_1, F_2\}$), and a sphere is not a torus. The angular-momentum triple of [Example 4.4](#) fails the same way, $X_{\ell_2}\ell_1 = \ell_3$.
- *Independence on Σ (I).* The pendulum $H = \frac{1}{2}p^2 - \cos \theta$ of (1.2), at $E = 1$. The level set is the separatrix, $p = \pm 2 \cos(\theta/2)$: two loops round the cylinder meeting at the unstable equilibrium $(\pi, 0)$, where $dH = 0$. It is compact and connected, and it is a figure-eight, not a circle.
- *Compactness (C).* The free particle $H = \frac{1}{2}p^2$ on $\mathbb{R}^2$. At $E > 0$ each component of the level set is a line $p = \pm\sqrt{2E}$, and the motion on it is not periodic. This is $T^0 \times \mathbb{R}^1$, and it is the shape of the conclusion whenever compactness fails.
- *Connectedness.* The pendulum at $E > 1$: the level set has two components, $p > 0$ and $p < 0$, each a circle. $\mathcal{M}_f$ is two tori, not one; the theorem is applied to each component separately.

Compactness is the hypothesis most often forgotten and the one [Lecture 9](#) will fail: the open Toda chain passes (I) and involution and has non-compact level sets (§ 9.5).

4.4 The proof, in six stages

The proof turns n commuting flows into an action of the group $\mathbb{R}^n$ on Σ , and then identifies Σ with a quotient of $\mathbb{R}^n$. Groups acting on spaces have not appeared in the course so far, so stage (0) sets up the three words the proof uses.

Stage (0): an action of $\mathbb{R}^n$, its orbits and its stabiliser

Definition 4.9 (Action, orbit, stabiliser). An *action* of $\mathbb{R}^n$ on a set Σ is a family of maps $\Phi_t: \Sigma \rightarrow \Sigma$, $t \in \mathbb{R}^n$, with

$$\Phi_0 = \text{id}, \quad \Phi_{s+t} = \Phi_s \circ \Phi_t.$$

The *orbit* of x is $\{\Phi_t(x) : t \in \mathbb{R}^n\}$; the action is *transitive* if some (hence every) orbit is all of Σ . The *stabiliser* of x is $\Lambda_x = \{t : \Phi_t(x) = x\}$.

Lemma 4.10. Λ_x is a subgroup of $\mathbb{R}^n$, the same for every point of the orbit of x , and $t \bmod \Lambda_x \mapsto \Phi_t(x)$ is a bijection from $\mathbb{R}^n/\Lambda_x$ onto the orbit.

Proof. If $s, t \in \Lambda_x$ then $\Phi_{s+t}(x) = \Phi_s(\Phi_t(x)) = x$, and $\Phi_{-t}(x) = \Phi_{-t}(\Phi_t(x)) = x$. If $y = \Phi_s(x)$ and $t \in \Lambda_x$, then $\Phi_t(y) = \Phi_{t+s}(x) = \Phi_s(\Phi_t(x)) = y$ — here we used $s + t = t + s$: the group is abelian. Finally $\Phi_s(x) = \Phi_t(x)$ if and only if $\Phi_{s-t}(x) = x$, i.e. $s - t \in \Lambda_x$. ■

Two examples fix the words ([Figure 4.2](#)). The oscillator flow is an action of $\mathbb{R}$ on the plane; the orbit of a point other than the origin is a circle with stabiliser $(2\pi/\nu)\mathbb{Z}$, while the origin is its own orbit, with stabiliser all of $\mathbb{R}$. And on the torus $\mathbb{R}^2/2\pi\mathbb{Z}^2$ the flow along the direction $(1, \sqrt{2})$ is an action of $\mathbb{R}$ with stabiliser $\{0\}$: the orbit is a copy of $\mathbb{R}$ that never closes, winds densely (§ 4.5), and is still not the whole torus. One flow cannot fill a two-dimensional space; the proof uses n flows on an n -dimensional one.

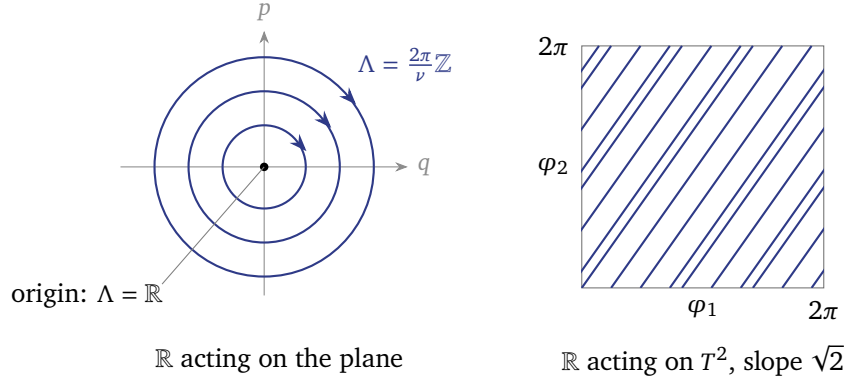


Figure 4.2: Orbits and stabilisers. Left: the oscillator flow, an action of $\mathbb{R}$ on the plane. Every orbit but one is a circle with stabiliser $\frac{2\pi}{\nu}\mathbb{Z}$; the origin is fixed, with stabiliser $\mathbb{R}$. Right: the flow of slope $\sqrt{2}$ on the torus (a square with opposite sides identified), drawn over ten crossings of the bottom edge. The stabiliser is $\{0\}$ and the orbit never closes, but a single orbit of $\mathbb{R}$ is a curve and cannot be the whole torus.

Stage (i): Σ is a Lagrangian submanifold

By (I) and the implicit function theorem, near each point of Σ the level set is a smooth connected piece of dimension $2n - n = n$, which therefore lies in Σ ; so Σ is a smooth n -dimensional submanifold with tangent spaces (4.2). The fields X_{F_j} are tangent to it, because

$$dF_i(X_{F_j}) = X_{F_j}F_i = \{F_i, F_j\} = 0. \quad (4.5)$$

At each point they are linearly independent, since $\iota_{X_{F_j}}\omega = dF_j$ and $v \mapsto \iota_v\omega$ is injective (3.1). So the n vectors $X_{F_1}, \dots, X_{F_n}$ are a basis of the n -dimensional tangent space of Σ at every point (Figure 4.3). On that basis $\omega(X_{F_i}, X_{F_j}) = \{F_i, F_j\} = 0$, so ω vanishes on $T\Sigma$: Σ is Lagrangian (Definition 3.21). *Spent here: independence, and involution twice.*

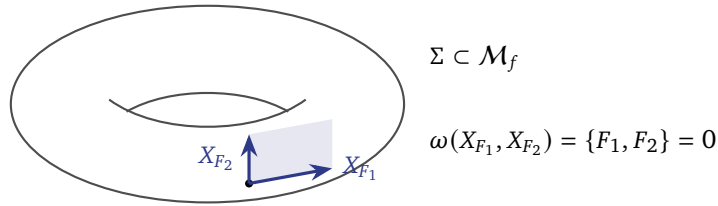


Figure 4.3: A regular compact level set with $n = 2$. The Hamiltonian vector fields of the integrals are tangent to it (4.5), independent, and therefore span each tangent plane; ω vanishes on the parallelogram they span. Stage (i) in one picture.

Stage (ii): n complete commuting flows, hence an action of $\mathbb{R}^n$

The fields X_{F_i} are tangent to Σ , so they restrict to vector fields on the compact manifold Σ , and by Proposition 1.8 they are complete there: each flow φ_s^i is defined on Σ for all $s \in \mathbb{R}$ and maps Σ to itself. By Corollary 1.16 these flows commute. Define

$$\Phi_t = \varphi_{t_1}^1 \circ \varphi_{t_2}^2 \circ \dots \circ \varphi_{t_n}^n, \quad t \in \mathbb{R}^n. \quad (4.6)$$

Then $\Phi_0 = \text{id}$, and $\Phi_{s+t} = \Phi_s \circ \Phi_t$ because the factors commute and each is a one-parameter group. This is an action of $\mathbb{R}^n$ on Σ . *Spent here*: compactness, for completeness — this is the promise [Lecture 1](#) made after [Proposition 1.8](#) — and involution, for commutativity.

Stage (iii): the action is transitive

Fix $x_0 \in \Sigma$ and consider the *orbit map* $\Phi^{x_0}: \mathbb{R}^n \rightarrow \Sigma$, $t \mapsto \Phi_t(x_0)$. Since the factors in (4.6) commute, any one of them may be put first, so

$$\frac{\partial}{\partial t_i} \Phi_t(x_0) = X_{F_i}(\Phi_t(x_0)).$$

The derivative of Φ^{x_0} at every t carries the standard basis of $\mathbb{R}^n$ to a basis of the tangent space (stage (i)), so by the inverse function theorem Φ^{x_0} is a *local diffeomorphism*: it maps a small ball about any t diffeomorphically onto a neighbourhood of $\Phi_t(x_0)$ ([Figure 4.4](#)).

Consequently every orbit is open in Σ . The orbits partition Σ , so the complement of one orbit is a union of others and is open too; an orbit is therefore open and closed. Σ is connected, so there is only one orbit. *Spent here*: independence (the local diffeomorphism), connectedness, and involution once more (the factors commute).

By [Lemma 4.10](#), with $\Lambda = \Lambda_{x_0}$ now the same for every point of Σ ,

$$\mathbb{R}^n / \Lambda \longrightarrow \Sigma, \quad t \bmod \Lambda \longmapsto \Phi_t(x_0) \text{ is a bijection.} \quad (4.7)$$

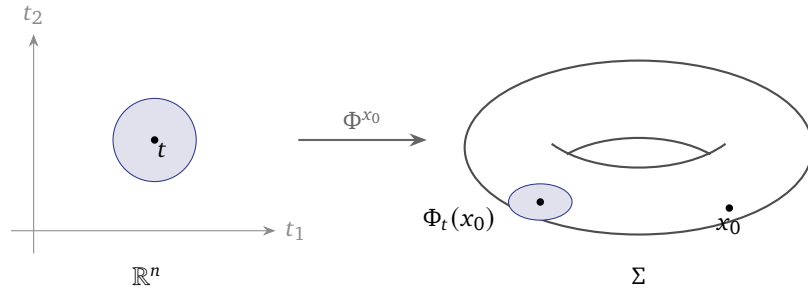


Figure 4.4: Stage (iii). Because the X_{F_i} are a basis everywhere, the orbit map carries a small ball about every t diffeomorphically onto a patch of Σ . So each orbit is open; in a connected Σ an open orbit whose complement is also open is everything. The map is not injective — that is stage (iv).

Stage (iv): Λ is a lattice of full rank

This is the step doing the most work, and it has two halves that must not be merged.

Λ is *discrete*. Choose a ball U about 0 on which Φ^{x_0} is injective (stage (iii)). If $\lambda \in \Lambda \cap U$ then $\Phi^{x_0}(\lambda) = x_0 = \Phi^{x_0}(0)$, so $\lambda = 0$. Translating, for any $\lambda \in \Lambda$ the ball $\lambda + U$ contains no other point of Λ , because $\Lambda - \lambda = \Lambda$. Every point of Λ is isolated. Contrast $\mathbb{Z} + \sqrt{2}\mathbb{Z} \subset \mathbb{R}$, a subgroup with points arbitrarily close to 0: what excludes that here is independence, through the local diffeomorphism.

Lemma 4.11 (Discrete subgroups of $\mathbb{R}^n$ · cited). A discrete subgroup of $\mathbb{R}^n$ is $\mathbb{Z}e_1 \oplus \cdots \oplus \mathbb{Z}e_k$ for some linearly independent $e_1, \dots, e_k$, $0 \leq k \leq n$. (Arnold, §49 D, Lemma 3.)

Discreteness alone gives a cylinder, not a torus. Complete $e_1, \dots, e_k$ to a basis $e_1, \dots, e_n$ of $\mathbb{R}^n$. Then

$$\mathbb{R}^k / \mathbb{Z}^k \times \mathbb{R}^{n-k} \longrightarrow \Sigma, \quad (\theta, y) \longmapsto \Phi^{x_0} \left(\sum_{i \leq k} \theta_i e_i + \sum_{j > k} y_j e_j \right)$$

is well defined and bijective by (4.7), and a local diffeomorphism by stage (iii). A bijective local diffeomorphism is a diffeomorphism. So $\Sigma \cong T^k \times \mathbb{R}^{n-k}$, and every k from 0 to n is possible for a discrete stabiliser.

Compactness forces $k = n$. $T^k \times \mathbb{R}^{n-k}$ is compact only if $k = n$, and Σ is compact. Hence Λ has full rank and $\Sigma \cong \mathbb{R}^n / \Lambda \cong T^n$. *Spent here*: independence for discreteness, compactness for full rank — the second use of compactness, and the one that fails for the open Toda chain.

Example 4.12 (Stage (iv) with $n = 1$). With one degree of freedom, Σ is a connected curve and $\mathbb{R}$ acts on it by the flow of H . The stabiliser is a discrete subgroup of $\mathbb{R}$, so either $\{0\}$ or $\tau\mathbb{Z}$ for some $\tau > 0$ (Figure 4.5). If it is $\tau\mathbb{Z}$ the orbit map wraps the line onto a circle, and τ is the period. If it is $\{0\}$, the orbit map is a bijection $\mathbb{R} \rightarrow \Sigma$: the free particle’s level line. Compactness is exactly what excludes the second case, and nothing else about the curve can: a compact connected curve without a zero of X_H is a closed orbit. (Stage (iii) is the easy half here: the orbit of x_0 is an open arc of Σ whose complement is open too, so it is all of Σ .)

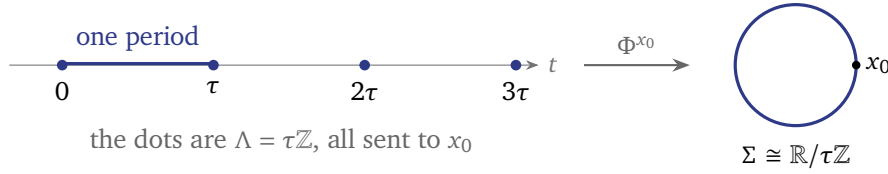


Figure 4.5: Stage (iv) with $n = 1$. The orbit map sends every point of the stabiliser $\Lambda = \tau\mathbb{Z}$ to x_0 and wraps each interval of length τ once round Σ . Discreteness allows $\Lambda = \{0\}$ as well, and then the line is not wrapped at all; only compactness rules that out.

Stage (v): the period lattice and a basis of cycles

Definition 4.13 (Period lattice). The stabiliser $\Lambda \subset \mathbb{R}^n$ of stage (iv) is the *period lattice* of Σ . A basis $e_1, \dots, e_n$ of it gives *basis cycles* $\gamma_i(s) = \Phi_{se_i}(x_0)$, $s \in [0, 1]$, closed loops on Σ .

A point $t \in \mathbb{R}^n$ comes back to x_0 exactly when $t \in \Lambda$, so the components of t in the basis e_i , times 2π , are angles. Write $A = (e_1 \cdots e_n)$ for the matrix whose columns are the basis vectors, and set

$$\varphi = 2\pi A^{-1}t \pmod{2\pi\mathbb{Z}^n}. \quad (4.8)$$

These are the angle coordinates of **Theorem 4.8**: $\varphi \mapsto \Phi_{A\varphi/2\pi}(x_0)$ identifies $\mathbb{R}^n / 2\pi\mathbb{Z}^n$ with Σ (Figure 4.6), and along γ_j the angle φ_j runs once from 0 to 2π while the others are constant.

Here is **Lecture 2**’s annulus again. Each φ_i is not a function on Σ , only a function modulo 2π , but $d\varphi_i$ is a well-defined 1-form, closed since it is locally d of a function. It is *not exact*:

$$\oint_{\gamma_j} d\varphi_i = 2\pi \delta_{ij} \neq 0 \quad (i = j),$$

whereas an exact form integrates to zero round every loop (Proposition 2.22). The torus carries n independent closed non-exact 1-forms, detected by n loops; Lecture 5 integrates $p \cdot dq$ over those same loops.

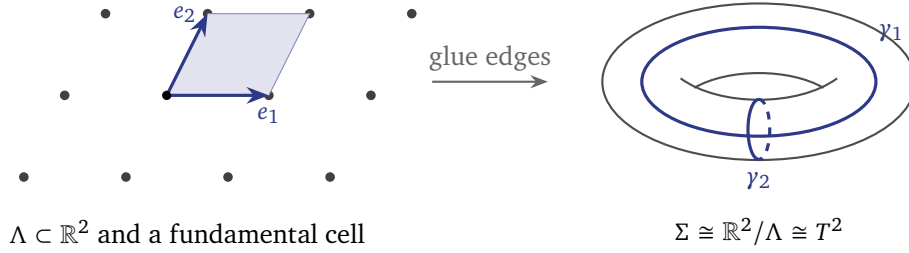


Figure 4.6: Stages (iv)–(v). Left: the period lattice Λ with a basis e_1, e_2 and the cell they span. Every point of $\mathbb{R}^2$ is a lattice translate of exactly one point of the cell, and gluing its opposite edges gives the torus. Right: the images of the edges, $t = se_1$ and $t = se_2$ for $s \in [0, 1]$, are the basis cycles γ_1, γ_2 — the cycles of Figure 2.12.

Stage (vi): straight-line motion, and what is left for Lecture 5

Let $H = h(F)$. On Σ the coefficients in (4.4) are the constants $c_i = \partial_i h(f)$, so $X_H = \sum_i c_i X_{F_i}$, whose flow is $\Phi_{\tau c}$: in $\mathbb{R}^n$ the motion is $t \mapsto t + \tau c$. By (4.8),

$$\varphi(\tau) = \varphi(0) + \nu \tau, \quad \nu = 2\pi A^{-1} \nabla h(f). \quad (4.9)$$

That is conclusion 3 of Theorem 4.8, and it completes the proof.

One more fact is needed next time, and it is a consequence of stage (i).

Proposition 4.14 ($p \cdot dq$ is closed on the torus). *If $\mathcal{M}$ is $\mathbb{R}^{2n}$ or a cotangent bundle, the restriction of $p \cdot dq$ to Σ is a closed 1-form.*

Proof. Restriction is pullback by the inclusion, and d commutes with pullback (2.15). So $d(p \cdot dq|_{\Sigma}) = (d(p \cdot dq))|_{\Sigma} = -\omega|_{\Sigma} = 0$, by Theorem 3.28 and stage (i). ■

So $\oint_{\gamma} p \cdot dq$ over a loop on Σ is unchanged when the loop is deformed within Σ (Corollary 2.26). That is what Lecture 5 needs to define the *action variables* $I_i = \frac{1}{2\pi} \oint_{\gamma_i} p \cdot dq$ and to show that, as f varies, the angles (4.8) and the I_i together form canonical coordinates on a whole neighbourhood of Σ . The theorem as proved here is about one torus at a time; how the tori fit together as f moves is Lecture 5's business, and the reading section on Hamiltonian monodromy at the end of this lecture shows it can be subtle.

Tutorial

One torus, two lattices

Example 4.15 (One torus, two lattices). Return to Example 4.3. The flow of X_{H_i} turns the i -th (q_i, p_i) -plane clockwise at angular speed ν_i and leaves the other alone.

- With $F = (H_1, H_2)$, Φ_t turns the planes through $\nu_1 t_1$ and $\nu_2 t_2$, so $\Lambda = \frac{2\pi}{\nu_1} \mathbb{Z} \oplus \frac{2\pi}{\nu_2} \mathbb{Z}$, a rectangular lattice.

- With $F = (H, H_2)$, X_H turns both planes and X_{H_2} the second, so Φ_t turns them through $\nu_1 t_1$ and $\nu_2(t_1 + t_2)$. Now $t \in \Lambda$ when $\nu_1 t_1$ and $\nu_2(t_1 + t_2)$ are both in $2\pi\mathbb{Z}$, and a basis is

$$e_1 = \frac{2\pi}{\nu_1}(1, -1), \quad e_2 = \frac{2\pi}{\nu_2}(0, 1).$$

Unless ν_2/ν_1 is an integer, $(2\pi/\nu_1, 0) \notin \Lambda$: the lattice is not the product lattice of the first bullet (Figure 4.7).

The level set is the same torus in both cases, and the two lattices have the same cell area $4\pi^2/(\nu_1\nu_2)$; they describe the same torus with different generators of the action. The basis cycles are the same too: in both cases γ_1 goes once round the first plane and γ_2 once round the second.

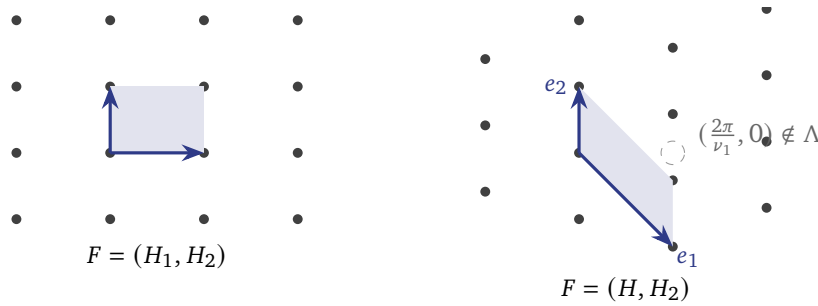


Figure 4.7: The period lattice of one oscillator torus, $\nu_1 = 1$, $\nu_2 = \sqrt{2}$, for two choices of integrals. Left: rectangular. Right: the lattice of Example 4.15, a different lattice with the same cell area. The lattice records the generators as well as the torus; the frequency vector read off it in stage (vi) does not depend on the choice.

By (4.9), with $H = H_1 + H_2$: the product lattice and $\nabla h = (1, 1)$ give $\nu = (\nu_1, \nu_2)$; the second lattice and $H = F_1$, $\nabla h = (1, 0)$, give $\nu = (\nu_1, \nu_2)$ again. The lattice depends on the choice of integrals; the motion does not.

Where each hypothesis was spent

Hypothesis	Stage	What it bought
Involution	(i), (ii), (iii)	tangency, Lagrangian; commuting flows; $\partial_{t_i}\Phi_t$ at every t
Independence on Σ	(i), (iii), (iv)	a manifold; local diffeomorphism; discrete Λ
Compactness	(ii), (iv)	complete flows; full rank of Λ
Connectedness	(iii)	one orbit

4.5 Quasi-periodic motion and resonance

The theorem says the motion is a straight line on a torus. It does *not* say the motion is periodic, and the difference is the subject of Lecture 6.

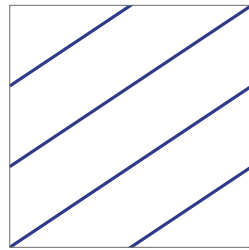
Definition 4.16 (Resonance). A frequency vector $\nu \in \mathbb{R}^n$ is *resonant* if $k \cdot \nu = 0$ for some nonzero $k \in \mathbb{Z}^n$, and *non-resonant* otherwise. A torus is resonant if its frequency vector is.

A change of lattice basis replaces ν by $B\nu$ with B an integer matrix of determinant ± 1 , and $k \cdot \nu = 0$ becomes $(B^{-T}k) \cdot (B\nu) = 0$: resonance is a property of the torus, not of the basis. For $n = 2$ it says exactly that ν_1/ν_2 is rational (or a frequency vanishes).

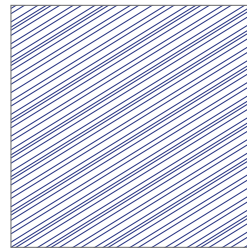
Proposition 4.17 (Winding on T^2). *Let $\varphi(t) = \varphi(0) + \nu t$ on T^2 with $\nu_2 \neq 0$. If ν_1/ν_2 is rational every orbit is closed; if it is irrational every orbit is dense.*

Proof (★). Watch the orbit cross the circle $\varphi_2 = 0$. It returns every $2\pi/\nu_2$, having advanced φ_1 by $2\pi\alpha$, $\alpha = \nu_1/\nu_2$; so the crossings are $\varphi_1^{(m)} = \varphi_1^{(0)} + 2\pi m\alpha \bmod 2\pi$. If $\alpha = a/b$ in lowest terms, the b -th crossing is the first, and the orbit closes. If α is irrational the crossings are all distinct (else $m\alpha \in \mathbb{Z}$ for some $m \neq 0$). Of any $N+1$ of them, two lie within $2\pi/N$ of each other on the circle; if they are crossings m and $m+d$, then every d returns rotate the circle by a nonzero angle smaller than $2\pi/N$, and iterating that rotation lands within $2\pi/N$ of any given point. So the crossings are dense in the circle. Every point of T^2 is reached from the circle by flowing for a time less than $2\pi/|\nu_2|$, and flowing for a fixed time is continuous, so the orbit is dense in T^2 . ■

For general n the same is true and harder: if ν is non-resonant every orbit is dense in T^n , and time averages of continuous functions along it equal their averages over the torus; if there are exactly r independent integer relations $k \cdot \nu = 0$, the closure of every orbit is a torus of dimension $n - r$ (cited: Arnold, §51 B, Corollary 1, and §51 D). *So a torus is not an orbit*: on a non-resonant torus a single trajectory visits every neighbourhood and never returns exactly, and only when $r = n - 1$ (and $\nu \neq 0$) is the torus foliated by closed orbits (Figure 4.8).



$\nu_2/\nu_1 = 2/3$: closed



$\nu_2/\nu_1 = (\sqrt{5} - 1)/2$: dense

Figure 4.8: Straight-line motion on T^2 , drawn on the square $[0, 2\pi]^2$ with opposite sides identified. Left: a rational ratio; the orbit closes after three turns of φ_1 . Right: the golden ratio, drawn over a long but finite time; continued forever the orbit comes arbitrarily close to every point and never closes (Proposition 4.17).

In the oscillator every torus has the same $\nu = (\nu_1, \nu_2)$, so they are all resonant or all not. That is special. Generically ν depends on f , resonant and non-resonant tori are interleaved, and Lecture 6 asks which of them survive a perturbation.

4.6 What the theorem does not give

Theorem 4.8 starts from n integrals in involution and says nothing about how to find them. For $n = 1$ the energy suffices (given $dH \neq 0$ on a dense open set), which is why every one-degree-of-freedom system is integrable. From $n = 2$ on, a second integral is an extra conserved quantity with no reason to exist, and generically it does not: Lecture 6 makes

that precise. Symmetry supplies some integrals, as it did for the central force, but the integrable systems of Part I — the Toda chain, Calogero–Moser — have far more integrals than symmetries, and [Lecture 7](#) gives the method that produces them: Lax pairs.

Reading

4.7 The theorem is local: Hamiltonian monodromy

[Theorem 4.8](#) describes one regular torus, and [Lecture 5](#) will extend its angle coordinates to a neighbourhood. Can the tori over *all* regular values be given coordinates at once? Not always. The obstruction is called *Hamiltonian monodromy*; it has nothing to do with the monodromy matrix of [Lecture 8](#), which shares only the name.

The spherical pendulum

A unit mass on a sphere of unit radius, in unit gravity, with polar angle θ from the upward vertical, azimuth ϕ , and $z = \cos \theta$ its height:

$$H = \frac{1}{2}p_\theta^2 + \frac{p_\phi^2}{2\sin^2 \theta} + \cos \theta, \quad J = p_\phi.$$

J , the angular momentum about the vertical, commutes with H by rotational symmetry, and $n = 2$. The level sets are compact, since $|p|^2 \leq 2(H + 1)$. At fixed $J = j$ the height moves in the reduced potential $V_j(z) = j^2/2(1 - z^2) + z$, whose minima are the circular conical motions: $j^2 = -(1 - z^2)^2/z$ with energy $E = (3z^2 - 1)/2z$, for $-1 < z < 0$. These curves bound the image of the *energy–momentum map* (H, J) ([Figure 4.9](#)). Over each point of these curves the fibre is a single circle; over their common endpoint $(-1, 0)$, the stable equilibrium, it is a point.

The unstable equilibrium at the top, $z = 1$, $p = 0$, is the other critical point. There $dH = dJ = 0$, it has energy 1, and its image $(1, 0)$ is an *isolated critical value* in the middle of the image. Every other interior point is a regular value with a compact connected fibre (Cushman–Bates, ch. IV, §3.4), a torus by [Theorem 4.8](#). The fibre over $(1, 0)$ contains the equilibrium and the trajectories that leave it and return to it; it is a torus with one cycle pinched to a point ([Figure 4.10](#)). So the set of regular values is a region with a puncture, and a loop round the puncture cannot be shrunk.

The period lattice, carried round the loop

Over each regular value there is a period lattice, and it varies continuously. Choose a basis at one point of the loop Γ and carry it continuously round Γ . It comes back as a basis of the same lattice, so the two differ by an integer matrix of determinant ± 1 . Take the basis $[\delta], [\gamma]$, with δ the J -orbit and γ the cycle made by flowing along H for the first-return time to the same J -orbit and then back along that J -orbit to the starting point. Then carried once round Γ , positively oriented, the basis returns as

$$\begin{pmatrix} [\delta] \\ [\gamma] \end{pmatrix} \mapsto \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} [\delta] \\ [\gamma] \end{pmatrix}, \quad (4.10)$$

(*cited*: Duistermaat 1980; the matrix, basis and orientation from Cushman–Bates, ch. IV, §5.2). The J -cycle comes back to itself — it is defined everywhere by the symmetry — and the other cycle comes back with a copy of it subtracted ([Figure 4.11](#)). Other sources print

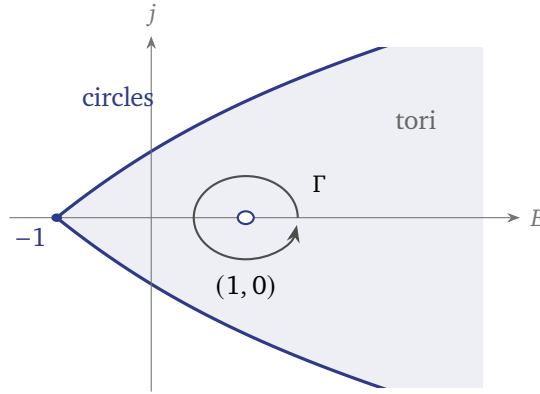


Figure 4.9: The image of the energy–momentum map of the spherical pendulum, drawn from the formulae in the text. The boundary is the curve of conical motions, over whose points the fibre is a circle; it meets the axis at the stable equilibrium $(-1, 0)$. The point $(1, 0)$, the unstable equilibrium, is an isolated critical value: the regular values form a punctured region, and the loop Γ cannot be contracted in it.

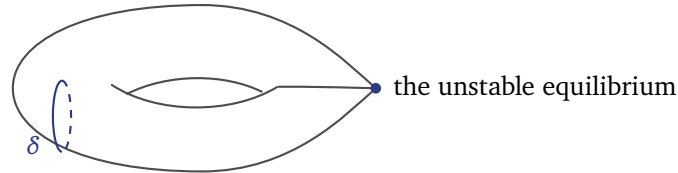


Figure 4.10: The fibre over $(1, 0)$: a pinched torus. The cycle δ , drawn where the tube is still fat, shrinks to a point at the pinch, which is the unstable equilibrium. On nearby tori δ is the orbit of the rotation generated by J , of period 2π . (Cushman–Bates, ch. IV, fig. 3.4.2.)

$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ for the same system: the entries depend on the basis and the orientation of the loop. The invariant content is that the matrix is a non-trivial shear, and it cannot be undone by any other choice of basis. One way to see it numerically: the *rotation number*, the advance of ϕ in one period of z divided by 2π , continued once round Γ , comes back changed by exactly 1.¹

What it means, and why it does not contradict the theorem

The theorem is about one torus and, via [Lecture 5](#), a neighbourhood of one torus, where the lattice basis can always be chosen continuously. Monodromy is about the whole family: if angle coordinates existed on all the tori over the punctured region at once, the basis they define would be carried round Γ unchanged, and (4.10) says it is not. So *the spherical pendulum has no global action–angle coordinates* (Duistermaat 1980). The same happens round any *focus–focus* equilibrium: one at which the linearised pair (H, J) is, in suitable canonical coordinates, the pair $(xp_x + yp_y, xp_y - yp_x)$ — a hyperbolic scaling and a rotation — provided its critical value is isolated and the nearby fibres, and the singular one, are compact and connected. The top of the sphere is such a point (*cited*: the geometric monodromy

¹`scripts/check-l4-algebra.py`, check 9 — a numerical check, not a proof.

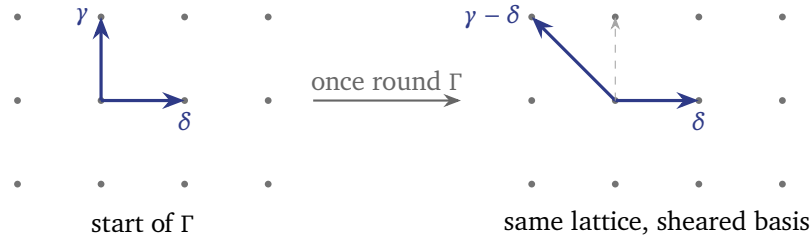


Figure 4.11: Hamiltonian monodromy. The period lattice returns to itself after one circuit of the puncture, but a basis carried continuously does not return to itself: δ is fixed and γ becomes $\gamma - \delta$, the matrix (4.10).

theorem, Cushman–Bates, ch. X, §2). [Theorem 4.8](#) proves the single-torus statement and [Lecture 5](#) the neighbourhood one; global triviality is false.

The quantum sequel. Quantised, H and J commute and have a joint spectrum, a set of points in the (E, j) -plane that is locally a deformed square lattice — that is Bohr–Sommerfeld, [Lecture 15](#). Carry a unit cell of it round $(1, 0)$ and it comes back sheared by the same matrix: the joint spectrum has a lattice defect at the puncture that no global quantum numbers can remove (*cited*: Cushman–Duistermaat 1988).

Summary.

- *The definition* is algebraic: n independent functions in involution on a $2n$ -dimensional phase space, one of them H ([Definition 4.6](#)). Each condition is cheap without the other ([Examples 4.4](#) and [4.5](#)), and compactness is not part of it.
- *The theorem* ([Theorem 4.8](#)): at a regular value, a compact connected component of the level set is a Lagrangian torus T^n , and the motion on it is $\varphi(t) = \varphi(0) + \nu t$.
- *The proof*: tangent commuting flows (the engine, [Corollary 1.16](#)), complete by compactness, give an action of $\mathbb{R}^n$; independence and connectedness make it transitive with discrete stabiliser Λ ; the quotient is $T^k \times \mathbb{R}^{n-k}$, and compactness forces $k = n$. The table in § 4.4 says where each hypothesis is spent.
- *The period lattice* Λ gives angles $\varphi = 2\pi A^{-1}t$, basis cycles γ_i and frequencies $\nu = 2\pi A^{-1}\nabla h$ (4.9). $p \cdot dq$ is closed on the torus ([Proposition 4.14](#)), which [Lecture 5](#) needs.
- *Resonance*: a torus is not an orbit. Rational frequency ratios give closed orbits, irrational ones dense orbits ([Proposition 4.17](#)).
- *The theorem is local* (reading): the spherical pendulum’s tori twist round the unstable equilibrium, and there are no global action–angle variables (the reading section on Hamiltonian monodromy).

Tutorial. Exercise Sheet 4.

References

Arnold, *Mathematical Methods of Classical Mechanics*, §49 (the theorem and its proof; Lemma 2 of §49 C is our stage (iii), Lemma 3 of §49 D our [Lemma 4.11](#)) and §51 (conditionally periodic motion, averaging, resonances); Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, ch. 2 (the theorem, and the examples); Cushman and Bates,

Global Aspects of Classical Integrable Systems, ch. IV (the spherical pendulum: §3.4 the fibres, §5 monodromy); J. J. Duistermaat, “On global action-angle coordinates”, *Comm. Pure Appl. Math.* 33 (1980); R. H. Cushman and J. J. Duistermaat, “The quantum mechanical spherical pendulum”, *Bull. Amer. Math. Soc.* 19 (1988). **Conventions.** Arnold and Bolsinov–Fomenko write $\omega = \sum dp \wedge dq$, the negative of ours; nothing in this lecture depends on the sign of ω , since every use of it is a statement that something vanishes.